Econ574 TermPaper

“Externality of Stigmatized Property”

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*Stigmatized Property (SP) :
Housing property that experienced undesirable event such as Murder, Suicide and so on.

- Japanese Law
  - If there was a stigmatized event in a transactional property, landlord has to disclose the fact to renter.
  - But not necessary to do so if a transactional property didn't experience such an event, no matter if there is a SP in the same apartment building.
  - If landlord can successfully hide the fact to others → No Externality.
- Paper Questions
  - Is there an externality of stigmatized event on other housing parcels in the same building?
  - Is there a spatial externality on neighborhood?
  - What determines the intensity of the externality?

- Data (Tokyo, Japan)
  - Stigmatized property
    - From *Jikobukken.Com* (http://www.jikobukken.com/)
    - 1,035 samples (based on reported events by ppl and media)
    - address, type of event, the date of event
      1. Discovery of dead body (189 cases)
      2. Death by fire (308)
      3. Death by unintentional accident (85)
      4. Suicide (193)
      5. Murder (260)
  - Rental housing
    - From *DoorChintai*, a realtor (http://chintai.door.ac/)
    - over 100,000 samples
    - address, rental price, housing characteristics
      - No sample matched to any stigmatized property
      - 140 samples matched to address of SP building → *Bld*
      - about 1,000 samples matched to block-level address of SPs → *Block*

- Method
  - Hedonic rental price function

$$\ln(Rent/m2)_i = \alpha_0 + \alpha_1 \ln(MinDist + 1)_i + \alpha_2 Bld_i + \alpha_3 Block_i + X_i \alpha_4 + \varepsilon_i$$

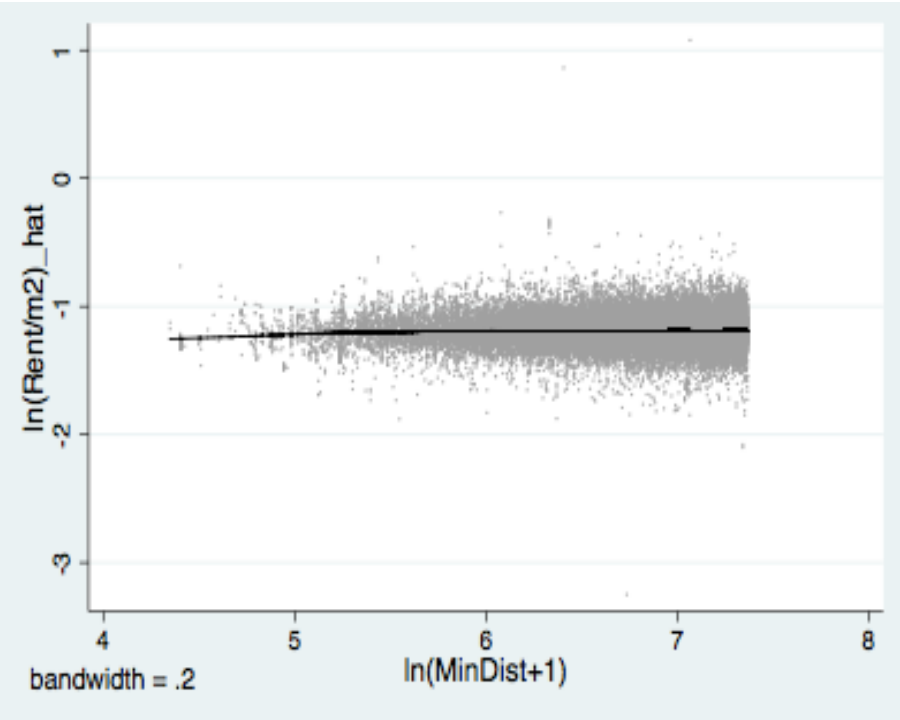
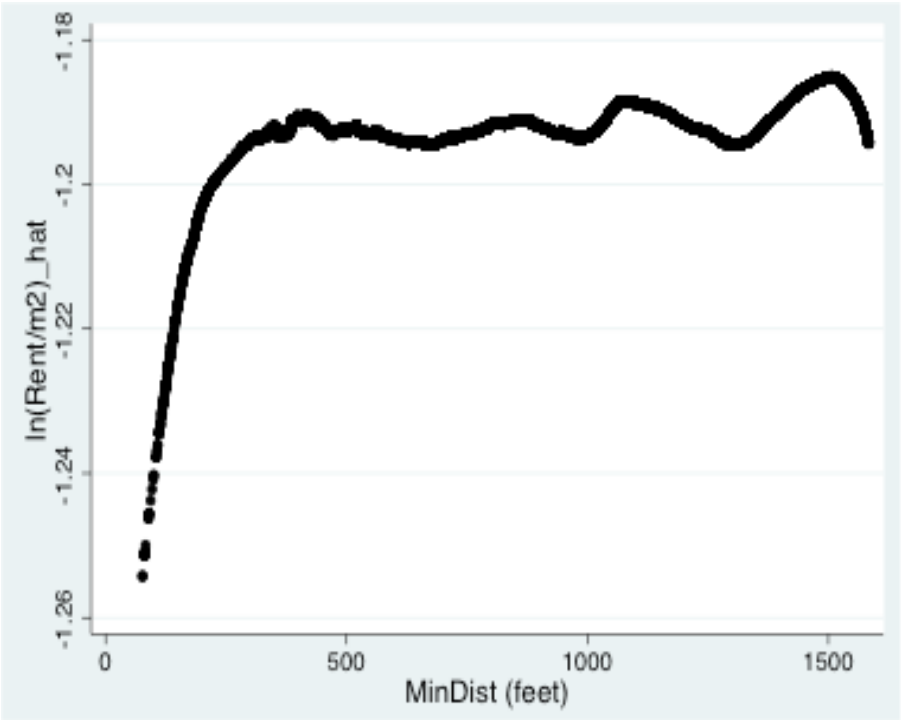
    - $MinDist_i$: Distance from rental housing *i* to its closest SP in feet, computed based on block-level geocodes.
    - Bld_i / $Block_i$: Dummy variables indicating a housing *i* located in building / block-area, which has a SP.
    - Control variables X_i : floor area, floor level, age, # bedrooms, distance to closest station, building structures, building types, closest stations, etc.
  - Targeted samples
    - The externality might be limited to a certain range from SP.
    - I run regressions using samples within 0.1 mile, 0.3 miles and 0.5 miles from SPs.
  - Additional model
    - Cross terms between interested variables and dummies about types of event

	0.1 mile	0.3 miles	0.5 miles	0.1 mile	0.3 miles	0.5 miles
ln(MinDist+1)	0.016***	0.003**	0.001			
×*De_Discovery*				0.017***	0.003**	0.002**
×*De_Fire*				0.018***	0.003**	0.001
×*De_Accident*				0.019***	0.003*	0.002**
×*De_Suicide*				0.016***	0.003**	0.001
×*De_Murder*				0.016***	0.003**	0.001
Bld	-0.027**	-0.026**	-0.028**			
×*De_Discovery*				-0.031	-0.022	-0.031
×*De_Fire*				-0.018	-0.027	-0.034*
×*De_Accident*				0.098	0.082**	0.077*
×*De_Suicide*				-0.028	0.015	0.021
×*De_Murder*				-0.047***	-0.050***	-0.049***
Block	0.105***	0.018*	0.010			
×*De_Discovery*				0.110***	0.023	0.014
×*De_Fire*				0.110***	0.019*	0.011
×*De_Accident*				0.102***	0.040**	0.041**
×*De_Suicide*				0.110***	-0.016	-0.028**
×*De_Murder*				0.111***	0.032***	0.022**
Observation	9,035	47,711	73,468	9,035	47,711	73,468
R-squared	0.815	0.793	0.797	0.816	0.793	0.797

Dependent variable = *ln(Rent/m2)* *** p<0.01, ** p<0.5, *p<0.1

- Semi-parametric estimates

$$\ln\,(Rent/m2)_i = \delta_0 + f(\ln\,(MinDist + 1)_i) + X_i\delta_4 + u_i$$

  - Partially linear regression model
  - Exclude samples whose *MinDist* is zero

e.g.) with samples within 0.3 miles from SPs ;